David Cropper

Melbourne, VIC · Australian Citizen

djcropster@gmail.com · 0400 811 545

LinkedIn: linkedin.com/in/david-cropper · Portfolio: journey.chaseriskandcompliance.com.au · GitHub: github.com/dcrops

AI Engineer

Professional Summary

Experienced engineer with 10+ years across testing, automation, data engineering and applied AI engineering.

Applies software engineering discipline to build end-to-end AI applications with **Python, FastAPI, Docker, and OpenAI APIs**, with independent AI Engineering portfolio work across retrieval systems, operational intelligence, explainable AI, and enterprise decision support.

Applies a disciplined **AI Engineering methodology** — architecture-first design, evidence-based validation, and human-in-the-loop review — to build AI systems with traceable, reviewable outputs for operational decision-making.

Selected Engineering Highlights

- Designed and delivered a **portfolio of AI applications** combining deterministic logic with AI reasoning across RAG, operational intelligence, diagnostics, entitlements, and career decision support.
- Built modular service architectures with **FastAPI**, containerised services with **Docker**, and unit/regression suites in **PyTest**.
- Designed explainable, evidence-backed recommendation flows with traceable outputs rather than opaque model responses.
- Published architecture notes, testing approach, and working demonstrations through a public portfolio and GitHub repositories.

Technical Skills

AI Engineering: OpenAI APIs · LLM integration · LangChain · Embeddings · Vector databases (ChromaDB) · AI evaluation · Explainable AI · Human-in-the-loop **Software Engineering:** Python · FastAPI · Pydantic · REST APIs · Docker · Git · PyTest · Unit / regression / behavioural testing **Data Engineering:** SQL · Pandas · AWS · Databricks

Professional Experience

AI Engineer — Independent Research & Development — Chase Risk & Compliance

Dec 2025 – Present · Independent engineering · Melbourne, VIC

- Delivered a portfolio of AI applications spanning retrieval-augmented generation, operational intelligence, explainable diagnostics, and enterprise decision support.
- Applied **FastAPI**, **Docker**, **OpenAI APIs**, **Git**, and **PyTest** where each portfolio system required them.
- Designed architectures that separate deterministic business logic from AI reasoning components.
- Published engineering documentation and public demonstrations of working systems.

Professional Development — Transition to AI Engineering

Oct 2023 – Nov 2025

- Completed Data Engineering upskilling across **Microsoft Fabric, Databricks, Snowflake, dbt, and Azure Data Factory**.
- Advanced into AI Engineering study and portfolio delivery using **Python, OpenAI APIs, and LangChain**.
- Strengthened delivery discipline through testing, documentation, and milestone-driven project execution.

Data Engineer — nbn Australia

Mar 2020 – Oct 2023

- Developed and maintained enterprise data pipelines and operational reporting using **AWS, Python, SQL, and Apache NiFi**.
- Performed SQL transformations, data validation, and production support for geospatial and business-operational datasets.
- Delivered in Agile teams supporting reliable production data systems with Git and automated testing.

Featured AI Projects

Career Intelligence Copilot

Overview: Supports evidence-based career and application decisions for AI Engineering roles. Reviews job opportunities against a structured career profile, ranks fit, recommends application strategy, and drafts tailored CVs for human review — reducing repetitive search work while keeping decisions explainable.

Engineering Highlights:

- Structured career profile model and evidence-cited opportunity assessment

- Deterministic application-strategy recommendation and evidence-grounded CV generation pipeline
- Golden validation suites with GO / NO-GO release gates
- Human-in-the-loop review of generated outputs

Technology Stack: Python · Pydantic · OpenAI APIs · PyTest · Git

Operational Intelligence Copilot

Overview: Demonstrates operational intelligence capability for understanding business data. Combines reliable analytics with AI reasoning to surface anomalies and produce clear, evidence-backed executive insights that can be checked and trusted.

Engineering Highlights:

- Intelligent intent routing and dataset profiling
- Anomaly detection with regression-tested analytics workflows
- Explainable, evidence-backed executive reporting

Technology Stack: Python · FastAPI · OpenAI APIs · PyTest · Git

Governance-Aware Document Intelligence RAG

Overview: Allows organisations to answer questions from their own documents with responses that stay grounded in source material. Built for teams that need trustworthy AI answers over policies, procedures, and knowledge bases — with evaluation and explainability controls.

Engineering Highlights:

- Multi-format document ingestion and embedding generation
- Retrieval orchestration with grounding validation
- AI evaluation framework, telemetry, and explainability controls

Technology Stack: Python · FastAPI · Docker · LangChain · ChromaDB · OpenAI · PyTest · Git

Payroll Diagnostics Engine

Overview: Reviews payroll data against deterministic business rules to identify anomalies. Produces explainable findings and executive-ready reports so issues can be investigated with confidence.

Engineering Highlights:

- Deterministic rule engine for payroll anomaly detection
- Explainable diagnostics and automated validation
- Executive-ready operational reporting

Technology Stack: Python · Pandas · OpenPyXL · PyTest · Git

Public Holiday Entitlements Application

Overview: Determines Australian public-holiday entitlements from location and rules for HR and payroll teams. Combines geospatial logic with external APIs to produce compliance-oriented results.

Engineering Highlights:

- Geospatial entitlement logic and business rules
- External REST API integration
- Compliance-oriented reporting

Technology Stack: Python · REST APIs · Geospatial Processing · Git

Earlier Experience

Earlier commercial roles as **Test Analyst** and **Test Automation Engineer** at Bakers Delight, Console, and AccessHQ, covering enterprise system delivery, test automation frameworks, Agile delivery, and production support.

AI Engineering Methodology

Applies AI to **improve engineering quality and delivery discipline**, not to replace engineering judgement.

Planning: Product vision · Architecture-first design · Roadmaps and milestones · Acceptance criteria

Implementation: Controlled AI-assisted coding · Architecture-aligned standards · Iterative delivery

Quality: PyTest regression and behavioural testing · Evidence-based debugging · Golden validation suites

Review: Architecture and maintainability reviews · Human-in-the-loop validation

Delivery: Git · Living documentation · Documentation freeze · GO / NO-GO release decisions

Courses & Upskilling

- General Assembly — Data Science Immersive (2019): Python coursework including NLP and predictive modelling projects
- LangChain & LLM Application Development
- Microsoft Fabric, Databricks, Snowflake, dbt
- Azure Data Factory

Certifications

- AWS Certified Developer - Associate
- Databricks Certified Data Engineer Professional